

Intel and DataStax partner to enhance the performance of Apache Cassandra and Apache Pulsar

Real-time data startup DataStax has announced a strategic partnership with Intel to improve the performance of systems based on the Apache Cassandra database and the Apache Pulsar messaging and streaming technology on the most recent Intel Xeon Scalable processors. In addition to providing enterprise and cloud solutions based on these potent open source initiatives, the companies intend to make improvements to each project. DataStax has been chosen for the Intel Disruptor Initiative, in which Intel works with a limited group of cutting-edge businesses that are redefining innovation.

Largely scalable databases, such as Apache Cassandra, function best on machinery designed specifically to support database workloads. Through this partnership, Intel and DataStax are working together to provide Apache Cassandra with CPUs that can support this utilisation in order to contribute to its long-term success. Apache Cassandra and Apache Pulsar's strength, along with Intel's long-lasting processors, will pave the path for efficient database operations and quick access to crucial data.

The Intel-DataStax partnership will also concentrate on introducing the most recent Cassandra and Pulsar technologies to academic institutions around the world in order to speed up innovation and train the next generation of programmers and data scientists on these potent open source initiatives.

With well-known products like Astra DB, DataStax's cloud database built on Apache Cassandra, the company is already a significant supporter of the Cassandra and Pulsar open source projects. Astra DB makes it simple for customers to access and utilise all of Cassandra's capabilities without having to deal with the hassle of managing infrastructure. End-to-end security, global scalability, and streamlined operations are features of this serverless platform.

infrastructure, in a blog post. In order to create StableHLO, a portable ML compute operation set that serves as a portability layer across machine learning frameworks and compilers, OpenXLA will endeavour to isolate the compiler from TensorFlow.

Open RAN platform for simulation of 5G networks is now available

The first open source platform for end-to-end simulation of 5G wireless networks, powered and optimised by Open Radio Access Network (Open RAN) components and ns-O-RAN (Open Radio Access Network Simulator), has been made available to the public by researchers at the Institute for the Wireless Internet of Things (WIoT) at Northeastern University.



The O-RAN standards-compliant Near-RT RIC is interfaced with using the ns-O-RAN emulator, which combines ns-3 with O-RAN-compliant features including the E2 application protocol and service models. This project is the product of a partnership between Mavenir, a member of the WIoT Industry Consortium and the Open6G initiative led by WIoT, and the Institute for the Wireless Internet of Things (WIoT) of Northeastern University.

The widely used open source network simulator ns-3, which simulates all the layers of the RAN protocol stack as well as core network operations and mobile traffic applications, and its 5G module from the University of Padova and NYU, which enables the LTE-5G NR dual connectivity RAN protocol stack on ns-3, serve as the foundation for the ns-O-RAN simulator.

With the help of extensive modelling of the entire 3GPP RAN protocol stack and large-scale 5G simulations based on 3GPP channel models, ns-O-RAN has been conceived and developed to enable the integration of O-RAN software, such as Mavenir's Near-RT RIC. This enables RAN key performance metrics (KPMs) data collection at scale, in various simulated circumstances, and with various applications (e.g., multimedia streaming, web browsing, wireless virtual reality, etc). Two E2 service models (E2SM), E2SM-Key Performance Metrics (KPM) monitoring and E2SM-RAN Control (RC), which enable a closed-loop control, are implemented by ns-O-RAN and support an O-RAN-compliant E2 interface (for example, of traffic steering and mobility).

Dutch minister in favour of using open source for digital ID wallets

The Dutch minister for digitalization, Alexandra van Huffelen, recently spoke about the need for open source in promoting ID wallets as secure and user-friendly.